

AVIEMORE, United Kingdom

WMO: 030630

Lat: 57.2064N

Lon: 3.8283W

Elev: 748

StdP: 14.30

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.4	21.5	13.8	11.4	17.8	18.6	14.4	23.7	29.5	42.1	26.1	41.5	1.8	230	0.621

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.7	74.7	61.3	70.7	59.8	67.3	58.1	63.2	71.3	61.3	68.5	59.5	65.7	7.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.8	79.0	66.5	58.0	73.9	64.6	56.4	69.6	62.7	28.9	71.7	27.6	68.6	26.3	66.0	72.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
22.5	18.9	16.3	DB	9.7	80.4	7.2	4.2	4.6	83.4	0.4	85.9	-3.6	88.2	-8.8	91.2
			WB	9.1	66.3	7.1	2.2	4.0	67.9	-0.1	69.2	-4.1	70.5	-9.3	72.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.5	36.2	36.8	39.5	43.5	48.2	53.2	57.0	56.1	52.3	46.4	40.4	36.1
	DBStd	9.41	6.07	5.98	5.72	5.63	5.79	5.00	4.86	4.51	4.71	5.46	6.07	7.71
	HDD50	2349	427	370	329	206	104	24	2	4	27	136	290	430
	HDD65	7114	892	789	792	644	520	356	252	279	381	577	737	895
	CDD50	719	0	1	2	12	49	118	220	193	97	24	3	0
	CDD65	9	0	0	0	0	0	1	5	3	0	0	0	0
	CDH74	140	0	0	0	0	11	24	66	38	1	0	0	0
	CDH80	13	0	0	0	0	0	4	5	4	0	0	0	0
	WSAvg	6.4	7.6	7.9	7.6	6.7	5.9	5.6	5.2	5.0	5.7	6.4	6.6	6.8
	PrecAvg	38.5	4.4	2.8	2.8	2.3	2.6	2.5	2.7	3.1	2.8	3.9	4.1	4.3
(15) Precipitation	PrecMax	53.2	11.3	9.2	8.6	5.1	4.2	5.7	5.5	6.3	8.3	6.5	7.4	10.5
	PrecMin	26.5	0.3	0.3	0.5	0.4	0.7	0.7	1.0	1.1	0.8	0.8	1.0	0.8
	PrecStd	5.7	2.3	2.1	1.8	1.1	1.0	1.1	1.2	1.3	1.5	1.5	1.4	2.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.7	53.4	59.9	67.2	74.8	77.0	80.0	79.1	70.7	62.0	56.8	53.6
		MCWB	48.4	48.6	49.3	53.0	59.3	60.9	63.9	64.2	60.4	55.2	52.7	50.2
	2%	DB	49.0	50.9	53.7	60.8	68.8	70.8	75.4	72.0	66.4	59.2	53.6	51.3
		MCWB	46.2	47.1	47.6	50.6	55.7	59.0	62.2	61.4	58.4	54.0	50.0	48.3
	5%	DB	47.2	48.3	50.5	56.7	63.6	66.2	71.1	67.7	63.3	57.2	51.7	49.2
		MCWB	44.5	45.0	45.9	49.0	53.1	57.0	60.7	59.6	56.8	52.5	48.3	46.5
	10%	DB	45.0	46.0	48.3	53.6	59.4	62.7	67.0	64.7	60.7	55.0	49.8	46.9
		MCWB	42.4	42.7	44.4	47.2	51.4	55.2	58.8	57.7	55.0	51.1	46.6	44.2
	0.4%	WB	48.8	50.3	50.9	54.5	61.3	63.1	66.1	65.9	62.1	56.6	53.3	50.9
		MCDB	51.0	53.1	56.8	62.9	72.9	73.2	76.9	73.6	68.3	60.4	56.1	52.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	46.6	47.5	48.2	51.8	57.0	60.2	63.6	62.5	59.6	54.8	50.8	48.8
		MCDB	48.8	50.3	52.5	58.8	65.8	68.8	72.2	69.3	64.7	58.5	53.3	51.0
	5%	WB	44.7	45.3	46.3	49.6	54.5	57.9	61.7	60.5	57.5	53.3	48.6	46.6
		MCDB	47.1	48.2	50.2	56.0	62.0	64.8	69.1	66.8	62.2	56.6	51.4	49.1
	10%	WB	42.6	42.9	44.5	47.7	52.1	55.8	59.6	58.6	55.7	51.4	46.7	44.3
		MCDB	44.9	45.7	48.2	53.0	58.6	61.7	66.0	63.8	59.9	54.5	49.5	46.9
(35) Mean Daily Temperature Range	5% DB	MDBR	9.0	10.4	12.5	15.4	17.3	15.4	15.7	15.3	14.3	12.4	10.3	9.9
		MCDBR	9.7	11.6	16.1	21.6	25.9	22.5	23.8	20.5	18.5	14.7	11.8	11.1
		MCWBR	8.6	9.2	11.0	13.0	14.5	12.1	12.5	11.6	11.6	10.5	10.1	10.1
	5% WB	MCDBR	9.9	10.3	15.5	19.2	22.8	20.0	20.6	18.5	16.7	13.3	11.7	11.4
		MCWBR	9.1	8.9	11.1	12.1	13.4	11.6	11.7	11.2	11.4	10.6	10.4	10.6
(40) Clear-Sky Solar Irradiance	taub		0.295	0.318	0.343	0.363	0.366	0.363	0.365	0.352	0.338	0.324	0.297	0.280
	taud		2.244	2.335	2.327	2.330	2.367	2.399	2.399	2.443	2.488	2.462	2.323	2.184
	Ebn at Noon		181	232	257	270	276	278	275	271	259	229	181	149
	Edh at Noon		16	23	30	34	35	35	34	30	25	20	15	13
(44) All-Sky Solar Radiation	RadAvg		136	337	662	1090	1462	1433	1380	1091	764	405	179	96
	RadStd		19	35	71	102	93	129	124	74	62	41	16	12

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page